

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

INSTRUMENT CLUSTER (IC) - VIRTUAL DISPLAY

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Instrument Cluster (IC) - Virtual Display. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument

Cluster, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1009-51	Ignition Authorization - Not programmed	- Instrument cluster power or ground circuit open circuit, high resistance - Instrument cluster is not configured correctly - High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Instrument cluster internal failure	- Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the instrument cluster power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform routine - Write Security Target ID (7006) - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
B1009-87	Ignition Authorization - Missing message	- Instrument cluster power or ground circuit open circuit, high resistance - Central junction box power or ground circuit open circuit, high resistance - High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Instrument cluster internal	- Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). A report value of 0V will be recorded when an internal value of 6V and below occurs. A report value of 25.5V will be recorded if the internal voltage should exceed 20V - Refer to the electrical circuit diagrams and check the

		failure	instrument cluster power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the central junction box power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
B100D-64	Column lock Authorization - Signal plausibility failure - Unable to unlock (engine start) or lock steering column lock	NOTE: This DTC is set when an Electric Steering Column Lock (ESCL) unlock request is received when the engine speed is 400RPM or vehicle speed is <1Kph or either signal is missing - Instrument Cluster (IC) power or ground circuit open circuit, high resistance - ESCL control module power or ground circuit open circuit, high resistance	- Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). A report value of 0V will be recorded when an internal value of 6V and below occurs. A report value of 25.5V will be recorded if the internal voltage should exceed 20V - Refer to the electrical circuit diagrams and check the Instrument Cluster (IC) power (pin 1) and ground circuits (pin 9 and 15) for open circuit, high resistance - Refer to the electrical circuit diagrams and check the Electric Steering Column Lock (ESCL) control module power and ground circuits for open circuit, high resistance

		High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B100E-64	Video Input "A" - Signal plausibility failure - Possible loss of navigation display turn by turn on the Instrument Cluster (IC)	- Integrated Audio Module (IAM) power or ground circuit open circuit, high resistance - Low-Voltage Differential Signalling (LVDS) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Instrument Cluster (IC) internal failure	NOTE: If the Instrument Cluster (IC) module is to be replaced, the new module will have to be initialised - Refer to the electrical circuit diagrams and check the Integrated Audio Module (IAM) power C2ME03A-1 and ground C2ME03A-8 circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the Low-Voltage Differential Signalling (LVDS) circuits C2ME03F-1 to C2MC02I-1 and C2ME03F-2 to C2MC02I-2 for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test - Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuits C2MC02A-1 to S2DB28E and C2MC02A-2 to S2BD27E for short circuit to ground, short circuit to

			power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, clear the DTCs and retest ■ If the fault persists, install a new Instrument Cluster (IC) ■ Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
B1026-12	Steering Column Lock - Vehicle will not start or crank - Unable to unlock or lock steering column lock	■ Electric Steering Column Lock control (ESCL) circuit short circuit to power ■ ESCL ground circuit open circuit, high resistance	NOTE: Output is not driven if DTC B9026-63 is set ■ Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index. Rectify before continuing ■ Refer to the electrical circuit diagrams and check the Electric Steering Column Lock (ESCL) control circuit for short circuit to power ■ Only tested at point of driving output ■ 'Driving' refers to powering up the circuit ■ if grounding condition noted - no further driving takes place ■ Refer to the electrical circuit diagrams and check the ESCL ground circuit for open circuit, high resistance
B1026-63	Steering Column Lock - Vehicle will not start or crank - Unable to unlock or lock	■ Circuit/component protection time-out ■ Electric Steering Column Lock (ESCL)	

	steering column lock	ground connection short to battery counter has incremented to the threshold and has latched OFF to protect the hardware from damage ESCL ground circuit short circuit to power	NOTE: Global DTC clear will not clear the DTC unless the counter is reset - Check Electric Steering Column Lock (ESCL) ground connection to Instrument Cluster (IC) - Refer to the electrical circuit diagrams and check the ESCL ground circuit for short circuit to power - DID/PID 1E7D Counter Reset can be used to reset the counter to 0 and re-allow operation of ESCL switched ground - Each time the fault counter is reset to zero, the fault threshold is halved - This can be repeated until the fault counter reaches 1, after which the DTC can no longer be cleared
B108E-13	Display - Instrument Cluster (IC) display corrupt or non-functional	- Instrument Cluster (IC) internal failure - Virtual display circuit open circuit, high resistance - IC is awake and continuity circuit in display flex cable is open circuit	NOTE: If the module is to be replaced, the new module will have to be initialised - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new Instrument Cluster (IC) - Execute VIN Learn

			Diagnostic Routine, write Odometer reading from old IC into Odometer PID
B10B7-86	Rear Air Discharge Temperature - Signal invalid	- Instrument cluster internal failure - Virtual display temperature sensor signal invalid	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
B115C-7A	Transfer Fuel Pump - Fluid leak or seal failure - Fuel level indication is incorrect	NOTE: This DTC is currently calibrated out in Software and shall not be logged - Fuel level sensor fault - Jet pump blocked or leaking - No warning light is illuminated to indicate that the DTC is logged but the fault may cause illumination of a telltale (Low Fuel Warning) as part of normal functionality under these conditions - Suction port pipe blocked or leaking	NOTE: If additional fuel level related DTCs are also set, perform the relevant corrective action(s) first - Using the manufacturer approved diagnostic system, check the central junction box for fuel level related DTCs and refer to the relevant DTC index - Check the integrity of the jet pump - Check the integrity of the suction port pipe
B1264-13	Control Module Connector(s) Loose Or Disconnected - Circuit open	- Instrument cluster internal failure - Virtual display circuit open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
B1264-49	Control Module Connector(s) Loose Or Disconnected - Internal electronic failure	- Instrument cluster internal failure - Airbag warning indicator circuit open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster

B1264-95	Control Module Connector(s) Loose Or Disconnected - Incorrect Assembly	- Instrument cluster internal failure - Alarm warning indicator circuit open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
B12FE-09	Fan - Component failures	NOTE: Software supports this DTC but will not log DTC as fan and fan circuit not fitted Instrument cluster cooling fan fouling	Check the integrity of the instrument cluster cooling fan
B12FE-12	Fan - Circuit short to battery	NOTE: Software supports this DTC but will not log DTC as fan and fan circuit not fitted Instrument cluster cooling fan circuit short circuit to power	Refer to the electrical circuit diagrams and check the instrument cluster cooling fan circuit for short circuit to power
B12FE-13	Fan - Circuit open	NOTE: Software supports this DTC but will not log DTC as fan and fan circuit not fitted Instrument cluster cooling fan circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the instrument cluster cooling fan circuit for open circuit, high resistance
B12FE-	Fan -		Refer to the electrical circuit

63	Circuit/component protection time-out	NOTE: Software supports this DTC but will not log DTC as fan and fan circuit not fitted Instrument cluster cooling fan circuit short circuit to power	diagrams and check the instrument cluster cooling fan circuit for short circuit to power
B1A14-96	Restraints Control Module (RCM) Warning Lamp - Component internal failure - Airbag readiness indicator may be permanently on, permanently off or dim	- Instrument Cluster (IC) is not configured correctly - IC internal failure - Airbag warning indicator circuit open circuit, high resistance	NOTE: If the module is to be replaced, the new module will have to be initialised - With reference to the technical bulletins library and using the manufacturer approved diagnostic system, check and install latest relevant level of software to the Instrument Cluster (IC) module - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new IC - Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
B1A68-86	Ambient Temperature Sensor - Signal invalid - Reduced backlighting at reduced temperature		

		NOTE: No warning light is illuminated to indicate that the DTC is logged, but the fault may cause changes in display illumination as part of defined functionality under these conditions. Instrument Cluster (IC) may appear dim - IC internal failure - IC internal temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Media Oriented System Transport (MOST) Fibre Optic Transceiver (FOT) sensor broken - IC unable to display audio/infotainment information - Backlighting sensor broken - IC will approximate the display temperature for backlight protection	NOTES: - Read individual Instrument Cluster (IC) replacement conditions - If the module is to be replaced, the new module will have to be initialised - Using the manufacturer approved diagnostic system, clear the DTCs and retest. - If the DTC persists, and Audio Display is lost, install a new IC - If the DTC persists, Audio Display is not lost and the customer has complained of reduced backlight, install a new IC - If the DTC persists, Audio Display is not lost and the customer has not complained of reduced backlight, do not install a new IC - Ambient temperatures of the country/area the vehicle operates in may not cause the customer operational issues - If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
B1A85-96	Ambient Light Sensor - Component internal failure - Back illumination may be incorrect; too bright or too dim		

NOTE:

If this DTC is logged without any symptom from customer then it should be ignored

- Instrument Cluster (IC) does not have the latest level software installed
- IC internal failure
 - Instrument cluster internal light sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- IC is awake and any one fitted internal light sensor has failed

NOTES:

- Read individual Instrument Cluster (IC) replacement conditions
- If this DTC is logged without any symptom from customer then it should be ignored
- If the module is to be replaced, the new module will have to be initialised

- With reference to the technical bulletins library, check the Instrument Cluster (IC) has the latest level software installed. Install latest level software as required and retest
- If the DTC persists, and Audio Display is lost, install a new IC
- If the DTC persists, Audio Display is not lost and the customer has complained of reduced backlight, install a new IC
- If the DTC persists, Audio Display is not lost and the customer has not complained of reduced backlight, do not install a new IC
 - Ambient temperatures of the country/area the vehicle operates in may not cause the customer operational issues
- If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID

P0460-11	Fuel Level Sensor A Circuit - Circuit short to ground	Active fuel level sensor circuit short circuit to ground	Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the active fuel level sensor circuit for short circuit to ground
P0460-15	Fuel Level Sensor A Circuit - Circuit short to battery or open	Active fuel level sensor circuit short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the active fuel level sensor circuit for short circuit to power, open circuit, high resistance
P0607-4B	Control Module Performance - Over temperature - Sudden changes in fan noise and reduction of back illumination	NOTE: Software supports this DTC but will not log DTC as fan and fan circuit not fitted - Instrument cluster cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance - Instrument cluster internal failure	- Refer to the electrical circuit diagrams and check the instrument cluster cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new instrument cluster
P060A-08	Internal Control Module Monitoring Processor Performance - Bus signal/message failures - Instrument		

	Cluster (IC) may lock or be non-functional	NOTE: Internal communication errors are causing lock-ups and resets - Instrument Cluster (IC) internal failure - Internal communication failures	NOTES: - Please use the special application on the SDD for Instrument Cluster (IC) blank screen - If the module is to be replaced, the new module will have to be initialised - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new IC - Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
P0610-55	Control Module Vehicle Options Error - Not configured	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
P1346-11	Fuel Level Sensor B Circuit - Circuit short to ground - Fuel level indication is incorrect	Passive fuel level sensor B circuit short circuit to ground	- Using the manufacturer approved diagnostic system, check the Central Junction Box (CJB) for related DTCs and refer to the relevant DTC index. - Refer to the electrical circuit diagrams and check the passive fuel level sensor 'B' circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest.
P1346-	Fuel Level Sensor B	Passive fuel level sensor B	Using the manufacturer

15	Circuit - Circuit short to battery or open - Fuel level indication is incorrect	circuit short circuit to power, open circuit, high resistance	approved diagnostic system, check the Central Junction Box (CJB) for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the passive fuel level sensor 'B' circuit for short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest
U0001-87	High Speed CAN Communication Bus - Missing message - Some Instrument Cluster (IC) functions not available	- At least one High speed CAN module has set their network management frame to a Bus off condition - Missing message from another control module via the high speed CAN bus (powertrain)	- Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module - Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0010-88	Medium Speed CAN Communication Bus - Bus off	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open

			circuit, high resistance
U0011-82	Medium Speed CAN Communication Bus Performance - Alive / sequence counter incorrect / not updated	- Invalid data received from another control module via the medium speed CAN bus - Medium speed CAN failure	- Using the manufacturer approved diagnostic system, check and install the latest relevant level of software to the instrument cluster. Clear the DTC and retest - Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network - Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-87	Medium Speed CAN Communication Bus Performance - Missing message - Some Instrument Cluster (IC) functions not available	- At least one Medium speed CAN module has set their network management frame to a Bus off condition - Missing message from another control module via the medium speed CAN bus (comfort)	- Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. - Check the relevant control module for related DTCs and refer to the relevant DTC index
U0300-00	Instrument Cluster (IC) Module Software Incompatibility - Loss of cluster function	- Instrument Cluster (IC) is not configured correctly - Sustained incompatibility checks cause this IC module to be taken out of the network system.	- Using the manufacturer approved diagnostic system, re-configure the Instrument Cluster (IC) with the latest level software - Using the manufacturer approved diagnostic

		This DTC may affect other vehicle functions	system, check for related DTCs and refer to the relevant DTC index
U2013-02	Switch Pack - General signal failure - Driver cannot navigate Instrument Cluster (IC) menus	NOTE: No warning light is illuminated to indicate that the DTC is logged, but the fault may cause illumination of a telltale, display of a message or failure of an information display as part of defined functionality under these conditions Steering wheel left/right switchpack internal failure	- Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new Switch Pack
U210A-86	Temperature Sensor - Signal invalid - Changes in display illumination, either too bright or too dim	NOTE: No warning light is illuminated to indicate that the DTC is logged, but the fault may cause changes in display illumination as part of defined functionality under these conditions - Instrument Cluster (IC) internal failure - Media Oriented System Transport (MOST) Fibre Optic Transceiver (FOT) sensor broken - IC unable to display audio/infotainment information	NOTES: - Read individual Instrument Cluster (IC) replacement conditions - If the module is to be replaced, the new module will have to be initialised - Using the manufacturer approved diagnostic system, clear the DTCs and retest - If the DTC persists remains, and Audio Display is lost, install a new IC - If the DTC persists remains, Audio Display is not lost and the customer has

		▪ Backlighting sensor broken - IC will approximate the display temperature for backlight protection	complained of reduced backlight, install a new IC ▪ If the DTC persists, Audio Display is not lost and the customer has not complained of reduced backlight, do not install a new IC ▪ Ambient temperatures of the country/area the vehicle operates in may not cause the customer operational issues ▪ If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
U3000-46	Control Module - Calibration/parameter memory failure - Odometer reading on Instrument Cluster (IC) is missing or incorrect	▪ Only logged as part of a self-test routine (0202) ▪ Instrument Cluster (IC) internal failure ▪ This DTC indicates that the contents of the Non-volatile Memory (NVM) containing the Odometer have become corrupted/inconsistent, potentially due to internal hardware failure	NOTE: If the module is to be replaced, the new module will have to be initialised ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ If the fault persists, install a new Instrument Cluster (IC) ▪ If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
U3000-49	Control Module - Internal failure - Some Instrument Cluster (IC) functions are lost, incorrect or inconsistent with vehicle fitment	▪ Instrument Cluster (IC) internal failure ▪ This DTC indicates that the contents of one or more FLASH downloadable memory areas have become corrupted/inconsistent, potentially due to	NOTE: If the module is to be replaced, the new module will have to be initialised

		internal hardware failure	■ Using the manufacturer approved diagnostic system, clear the DTCs and retest ■ If the fault persists, using the manufacturer approved diagnostic system reprogram (reFLASH) the Instrument Cluster (IC) Central Configuration File (CCF): 'as built data' ■ If the fault persists or the software repeatedly fails to complete the download process, install a new IC ■ If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
U3000-55	Control Module - Not configured	■ Incorrect wheels/tires installed ■ Tire size compensation is incorrectly configured	■ Install the correct wheels and tires ■ Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
U3000-87	Control Module - Missing Message - Some Instrument Cluster (IC) functions are lost, incorrect or inconsistent with vehicle fitment Control Module	NOTE: No warning light is illuminated to indicate that the DTC is logged, but the fault may cause illumination of a telltale, display of a message or failure of an information display as part of defined functionality under these conditions ■ Central Configuration File (CCF) mismatch with	■ Refer to manufacturers wiring diagram and inspect 'CO' CAN harness MS CAN High (CO) C2MC01A-23 through to S2DB27X and MS CAN Low (CO) C2MC01A-7 through to S2DB28X ■ Refer to manufacturers wiring diagram and inspect 'CO' CAN harness between Instrument Panel Console (IC) to J1926 OBD11 - MS CAN High (CO) C2MC01A-23 through S2DB27X and MS CAN Low (CO) C2MC01A-7 through to S2DB28X ■ If wiring condition is good -

		vehicle specification ■ CAN harness circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, reprogram (re-flash) the Central Configuration File (CCF) 'as built data' ■ 'as built data' - Software detailing data distributed to modules to function as per specification/equipment fitted and vehicle market stored in one or more control modules on the vehicle
U3002-81	Vehicle Identification Number (VIN) - Invalid serial data received - Some security functions remain locked	■ Incorrect Central Junction Box (CJB) installed ■ Incorrect Instrument Cluster (IC) installed	NOTES: ■ If the Central Junction Box (CJB) module is to be replaced, the new module will have to be initialised ■ If the Instrument Cluster (IC) module is to be replaced, the new module will have to be initialised ■ Using the manufacturer approved diagnostic system, check Central Junction Box (CJB) has correct Vehicle Identification Number (VIN) locked in ■ If VIN incorrect, install the correct CJB ■ If a new CJB is to be installed, configure using Land Rover approved diagnostic equipment. ■ Using the manufacturer approved diagnostic system, check Instrument Cluster (IC) has correct VIN locked in

			▪ If VIN incorrect, Install the correct IC ▪ If new IC installed; Execute VIN Learn Diagnostic Routine, write Odometer reading from old IC into Odometer PID
U3003-16	Battery Voltage - Circuit voltage below threshold - Some Instrument Cluster (IC) functions not available	NOTE: Battery voltage below 9v for 1s during self-test diagnostic routine 0202 ▪ Instrument Cluster (IC) power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the instrument cluster power and ground circuits for open circuit, high resistance ▪ A report value of 0V will be recorded when an internal value of 6V and below occurs. A report value of 25.5V will be recorded if the internal voltage should exceed 20V ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3003-17	Battery Voltage - Circuit voltage above threshold - Some Instrument Cluster (IC) functions not available	NOTE: Battery voltage above 16v during self-test diagnostic routine 0202 ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). ▪ A report value of 0V will be recorded when an internal value of 6V and below occurs. A report value of 25.5V will be recorded if the internal voltage should exceed 20V ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
U3003-	Battery Voltage -	▪ Mismatch between the	▪ Using the manufacturer

62	Signal compare failure - Some Instrument Cluster (IC) functions not available	voltage at the Instrument Cluster (IC) and the voltage value broadcast on the CAN bus	approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111) - and compare it to battery voltage ▪ A report value of 0V will be recorded when an internal value of 6V and below occurs. A report value of 25.5V will be recorded if the internal voltage should exceed 20V ▪ Refer to the electrical circuit diagrams and check the Instrument Cluster (IC) power C2MC01A-1 and ground C2MC01A-9 circuits for open circuit, high resistance
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